

Gradients, Gradient Descent & Assessment

Mathematics Foundations — Session 4

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Today's Goal

- Read the gradient as the direction of steepest ascent
 - Code gradient descent one step at a time and see the effect of the learning rate
 - Locate minima with the first- and second-derivative tests, and see why a neural-network loss has no easy global minimum
- Applied check: a matrix product, a chain-rule derivative, one gradient-descent step, plus an individual oral question

Plan

1. Opening Quiz
2. The Gradient
3. Gradient Descent
4. Minima
5. Applied Check (Final Hour)
6. Recap
7. To Do for Next Session

Opening Quiz

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Opening Quiz: Derivatives & the Chain Rule

The Gradient

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The Gradient as a Vector of Partial Derivatives

Gradient Descent

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Gradient Descent

Minima

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Minima

Applied Check (Final Hour)

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Applied Check

Individual Oral Question

Deliverable

A working one-step gradient-descent snippet, and the completed applied check.

Recap

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Recap

To Do for Next Session

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To Do for Next Session
